

GENERAL THETA FOUNDATIONS I: COMPATIBLE POLYHEDRAL LIFTS AND QUANTITATIVE CAUSAL WIDTH

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ABSTRACT. An exact finite-horizon stochastic transducer determines a sequence of simplex-section extensions joined by stochastic maps. Static extension size does not certify those maps. We formulate the compatibility condition, give its linear dual for fixed extensions, and exhibit an affine map whose best stochastic extension has uniform error exactly $1/4$. For a fixed-alphabet rotation experiment of separate positive rank three, a mean-support budget improves the all-hidden-state lower bound to $\sum_{t < N} 2^{-3Kt} \leq 320$ at the golden angle. A resonant polygon construction improves the exact upper width from order $N^{1/2}$ to order $N^{1/3}$. More generally, for every irrational angle with bounded continued-fraction coefficients, the minimum width among vector-state realizations is of order $N^{1/3}$; the lower bound covers all time-varying polygon chains, not only regular polygons. For general hidden realizations the resulting bounds are logarithmic and cube-root, and need not coincide. An elementary denominator argument gives an explicit doubly logarithmic lower bound for the rational rotation $(3 + 4i)/5$. Finally, requiring a single clock-free decoder to answer at any stopping length up to N gives linear autonomous state complexity. These results distinguish static lifts, compatible clocked lifts, vector-state realizations, and autonomous devices without identifying their different optimization problems.

1. INTRODUCTION

The state count of a positive realization is not determined by its linear rank. This is a classical fact. Benvenuti and Farina constructed transfer functions of fixed McMillan degree three with arbitrarily large minimum positive order [1]; subsequent work obtained matching realizations for their family [5, Example 2]. There is a second, different classical fact: a polygon with many vertices may have an extension with only logarithmically many facets [14, 6, 12]. The simultaneous finite-horizon problem lies between these observations. Its state register supplies an extension at each epoch, but also supplies stochastic maps joining the extensions for every allowed command. Neither independent positive factorizations nor independent small extensions impose this compatibility.

We study the complete binary-output experiment

$$(1) \quad C_{N,\alpha}(b \mid x, c_1, \dots, c_N, j) = \frac{1 + b e_j^T R_\alpha^{c_1 + \dots + c_N} x / 10}{2},$$

where $x \in \{-1, 1\}^2$, $c_t \in \{0, 1\}$, $j \in \{1, 2\}$ and $b \in \{-1, 1\}$. Here R_α is rotation through $2\pi\alpha$. All alphabets are fixed. All probabilities lie in the fixed interval

$$\left[\frac{1 - \sqrt{2}/10}{2}, \frac{1 + \sqrt{2}/10}{2} \right].$$

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Equality is required for every word, not just in expectation over an input prior. The separate positive minimum at every internal cut is three (Proposition 2.3). The question is the peak number of labels needed by one compatible machine.

1.1. The quantitative distinction. Write $W_N(\alpha)$ for unrestricted clocked positive-realization width. Write $V_N(\alpha)$ for its vector-state restriction, in which every state has a predictive vector and each stochastic command row intertwines those vectors, not merely their reachable affine combinations. These are defined in Section 2. Let $A_N(\alpha)$ be the number of states in a clock-free machine with one fixed decoder that answers a terminating query at any command length from zero through N . The last task has a stronger query-time specification, which is part of its definition.

Theorem 1.1 (Quantitative width distinctions). *Let α be irrational, and suppose all its continued-fraction partial quotients are at most an integer $A \geq 1$. Let q_N be the first convergent denominator at least $\max\{5, (40N)^{1/3}\}$. For $N \geq 1$,*

$$(2) \quad \max\left\{3, \frac{1}{3} \log_2 \frac{N}{160(A+1)}\right\} \leq W_N(\alpha) \leq V_N(\alpha) \leq q_N,$$

$$(3) \quad \max\left\{3, \left(\frac{N}{160(A+1)}\right)^{1/3}\right\} \leq V_N(\alpha) < (A+1) \max\{5, (40N)^{1/3}\}.$$

Thus $V_N(\alpha) = \Theta_\alpha(N^{1/3})$. In addition, for every infinite-order planar rotation, without the bounded-type assumption,

$$(4) \quad N+1 \leq A_N(\alpha) \leq 3(N+1)+2.$$

At the golden angle $\alpha = (\sqrt{5}-1)/2$, every exact hidden-state profile satisfies $\sum_{t < N} 2^{-3K_t} \leq 320$; every vector-state profile satisfies $\sum_{t < N} K_t^{-3} \leq 320$.

The hidden-state lower bound is not replaced by the stronger vector-state bound. A section of a K -coordinate simplex can project to many more than K vertices. This distinction is indispensable: regular polygons have small static extensions, and an unproved restriction to vector states would discard precisely the possible hidden savings. Theorem 3.5 specifies the additional stochastic compatibility required of a sequence of extensions. Theorem 3.2 checks it by a finite linear program when the extensions and the proposed lifted affine maps have been supplied. It does not optimize over all extensions efficiently.

For the rational rotation $R = (1/5) \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}$, Theorem 6.2 gives the explicit necessary condition

$$N \leq 140 2^{2W_N} 5^{82^{W_N}}.$$

It requires neither quantifier elimination nor an unproved bounded-type property of $\arg(3+4i)/(2\pi)$. Its growth rate is deliberately separated from the sharper bounded-type statement.

1.2. Mechanism and scope. For an observable polygon P , use the mean support $m(P) = (2\pi)^{-1} \int h_P$, rather than its area. The one-step gain for $\text{conv}(P \cup R_\alpha P)$ accumulates linearly along a finite orbit. Comparing that orbit hull with the optimal perimeter of an inscribed polygon gives a gain of order M^{-3} for at most M vertices and bounded-type angles. No squaring of a Hausdorff displacement is needed. The matching vector-state upper bound aligns regular polygons with convergents of the angle; their required radial expansion per step is then of order M^{-3} rather than M^{-2} .

All exact clocked counts use atomic stochastic rows, an external epoch, and known read-only tables. They do not count arithmetic or table construction. The new autonomous comparison removes the external epoch and fixes the decoder, but still uses the same atomic-row convention.

The old rational fair-bit compiler remains a separately priced implementation theorem [15]. No table-free bit-space claim is made.

Item	Mathematical role
Reachable simplex sections	Inherited finite-horizon lift; classical accessible-space and extension principles.
Stochastic extension dual	Finite convex separation, used to distinguish static affine lifts from implementable maps.
Mean-support budget and resonance	New quantitative argument and exact clocked construction; matching order within the vector-state class.
Rational-angle rate	Explicit arithmetic consequence of the same budget.
Anytime autonomous comparison	Finite-horizon consequence of Cayley–Hamilton and the classical peripheral-spectrum constraint.

The complete earlier proofs, including the volume theorem and tagged profile composition, are preserved separately. They are not prerequisites for auditing the new argument, and their archive size is not part of its mathematical content.

2. MACHINES AND OBSERVABLE SECTIONS

We use row vectors for state distributions. Put $\Delta_k = \{e \in \mathbb{R}^k : e \geq 0, e\mathbf{1} = 1\}$, $S = [-1/10, 1/10]^2$, and $D = [-1, 1]^2$.

Definition 2.1 (Clocked realization). A realization of (1) consists of an initialization $E_x \in \Delta_{K_0}$, row-stochastic matrices $T_{t,c} \in \mathbb{R}_+^{K_t \times K_{t+1}}$ for $0 \leq t < N$, $c \in \{0, 1\}$, and decoder means $d_j \in [-1, 1]^{K_N}$, satisfying

$$(5) \quad E_x T_{0,c_1} \cdots T_{N-1,c_N} d_j = e_j^T R_\alpha^{\sum c_t} x / 10$$

for every input word and query. The decoder emits the binary law with this conditional mean. The available-label profile is $(K_t)_{t=0}^N$; its peak is $\max_t K_t$, and its minimum is $W_N(\alpha)$. A final retained binary output can be implemented in two further output labels at its own cut and does not change this peak, which is at least three.

Only the current register persists. The seed, old commands, random coins, and protocol history are not supplied again. The seed and terminal query are atomic inputs; commands are external, rather than selected by an experimental policy. A row may depend on the external epoch and current command, but not on a future symbol. Every row, including an unreachable one, is normalized. Available-label counting permits padding and makes feasible profiles upward closed. The number of program bits, exact row arithmetic, and sampling cost are not part of this width. For comparison, a separate minimum at cut t imposes no bound at any other cut. It is not the peak of a simultaneously minimal machine.

Definition 2.2 (Vector-state realization). A clocked realization is vector-state if there are vectors $z_{t,s} \in D$ satisfying

$$(6) \quad \sum_s E_x(s) z_{0,s} = x/10, \quad \sum_{s'} T_{t,c}(s, s') z_{t+1,s'} = R_\alpha^c z_{t,s}, \quad d_j(s) = e_j^T z_{N,s}.$$

The middle equality is required at every available state, not only after averaging over reachable distributions. The least peak in this subclass is $V_N(\alpha)$.

Such a machine is equivalent to a sequence of polygons $P_t = \text{conv}\{z_{t,s}\}$ with

$$(7) \quad S \subset P_0, \quad P_t \subset D, \quad \text{conv}(P_t \cup R_\alpha P_t) \subset P_{t+1}.$$

The forward implication follows from (6); conversely, choose the vertices as states and the corresponding barycentric rows. Thus the minimum vertex count of a compatible chain is exactly the vector-state optimization. No such identification is imposed on W_N .

Proposition 2.3 (Separate minimum). *Every internal cut of (1) has normalized residual rank three and separate positive minimum three, for every N and rotation.*

Proof. The four prefixes with all commands idle have predictive vectors $x/10$ and span an affine plane. The two idle-suffix coordinate queries separate this plane. All residual arrays are affine functions of a two-dimensional predictive vector, so their normalized rank is exactly three. The vectors have Euclidean norm $\sqrt{2}/10$. The triangle with vertices

$$(1/2, 0), \quad (-1/2, 1/2), \quad (-1/2, -1/2)$$

contains the disk of this radius: its three supporting lines have distances $1/2, 1/(2\sqrt{5}), 1/(2\sqrt{5})$ from zero, all greater than $\sqrt{2}/10$. Its vertices have norm at most $1/\sqrt{2}$. Consequently every vertex defines a legal entire future continuation under all rotations and both queries. Barycentric initialization at the selected cut gives a three-generator factorization of all its residuals. To embed it into a complete machine, store the raw prefix until the selected cut, sample that three-valued label, and then retain the label and the remaining finite command count. Other cuts are unrestricted. Rank supplies the matching lower bound. This is the inherited separate minimum calculation of [16], included to fix its quantifier. $\square$

Lemma 2.4 (Reachable-section lift). *Every realization determines affine sections $Q_t = H_t \cap \Delta_{K_t}$ and linear maps π_t with $P_t = \pi_t(Q_t)$ satisfying (7) and*

$$(8) \quad |\text{ext } P_t| \leq 2^{K_t}.$$

For vector-state realizations one may instead use $|\text{ext } P_t| \leq K_t$.

Proof. Let E_h denote the state distribution following an actual prefix, and set $H_t = \text{aff}\{E_h\}$. Define $\pi_t(e)$ by the two terminal means obtained from e after idling all remaining commands. It maps the whole simplex into D . The identities

$$E_h T_{t,c} = E_{hc}, \quad \pi_{t+1}(E_h T_{t,c}) = R_\alpha^c \pi_t(E_h)$$

extend affinely across H_t . Stochasticity then gives $Q_t T_{t,c} \subset Q_{t+1}$ and the required observable inclusions. The idle command implies $P_t \subset P_{t+1}$, hence every P_t contains S . In its affine hull Q_t has at most K_t coordinate inequalities. Distinct vertices have distinct active zero-coordinate sets: otherwise their difference is a two-sided feasible direction at either vertex. There are at most 2^{K_t} such sets, and projection cannot increase the number of vertices. This is the standard face-count bound for extensions [6], not a special new combinatorial principle. For the vector-state class, use the polygon in (7). $\square$

The restriction to H_t is essential. A hidden basis state need not have the observable intertwining identity. The proof therefore retains the section before projection, as in accessible-space and observable-quotient constructions [3, 10]. It is this lift that separates the two bounds in Lemma 2.4.

3. STATIC EXTENSIONS AND STOCHASTIC COMPATIBILITY

A polyhedral extension is a polytope followed by an affine projection. Its size is the number of inequalities, not the number of projected vertices. The distinction is classical [14, 6]. We first record exactly what simplex normalization does, and does not, provide.

Proposition 3.1 (Normalization of bounded extensions). *Every positive-dimensional bounded polytope with f facets is affinely isomorphic to a section $H \cap \Delta_f$. Thus a bounded extension with f facets can be put in simplex-section form without increasing f . An affine map between two such sections need not extend to a stochastic map between their ambient simplices.*

Proof. Work in the affine hull and write an irredundant facet description $\ell_i(u) = b_i - a_i \cdot u \geq 0$, $1 \leq i \leq f$. Boundedness implies that the normals positively span the dual space. Consequently there are strictly positive w_i with $\sum_i w_i a_i = 0$. For example, express each $-a_i$ as a nonnegative combination of the normals and add these f dependences. The positive constant $c = \sum_i w_i b_i$ can be evaluated at a relative interior point, where it is a sum of positive slacks. The affine map

$$u \mapsto (w_i \ell_i(u)/c)_{i=1}^f$$

is injective, has coordinate sum one, and sends the polytope onto the intersection of its affine image with Δ_f . Injectivity follows because the normals span. A point polytope uses Δ_1 . The last assertion is proved quantitatively in Proposition 3.3 below. $\square$

The projection of a normalized extension is defined on its affine section. It need not map the whole ambient simplex into the observation domain. Nor does normalization produce transition kernels. These two additional requirements are not part of static extension complexity.

Theorem 3.2 (Stochastic extension criterion). *Let $Q \subset \Delta_k$ be a polytope, let $q^1, \dots, q^r$ affinely span Q , and let $F : Q \rightarrow \Delta_l$ be affine. Put $E_{hi} = q_i^h$ and $B_{hj} = F(q^h)_j$. A row-stochastic k by l matrix T satisfying $eT = F(e)$ on Q exists if and only if, for every real r by l matrix Z ,*

$$(9) \quad \langle Z, B \rangle \leq \sum_{i=1}^k \max_{1 \leq j \leq l} (E^T Z)_{ij}.$$

For fixed rational E, B , feasibility and a separating certificate are finite linear-programming problems.

Proof. The desired set of image matrices is the compact convex polytope $\mathcal{B} = \{ET : T \geq 0, T\mathbf{1} = \mathbf{1}\}$. Its support function at Z is the right side of (9), since the rows of T are independent probability vectors. Finite-dimensional separation says $B \in \mathcal{B}$ exactly when all the displayed inequalities hold. Equality on the spanning rows extends to Q by affinity. This is the ordinary support-function dual of a stochastic post-processing problem; it is not a new minimax interchange. The additional point is that the same test can be applied to each proposed inter-epoch lift. $\square$

Proposition 3.3 (An exact extension defect). *On*

$$Q = \{e \in \Delta_4 : e_1 + e_2 = e_3 + e_4 = 1/2\}$$

the affine map $F(e) = (2e_1, 2e_2)$ takes values in Δ_2 , but has no ambient stochastic extension. More precisely,

$$(10) \quad \min_{T \geq 0, T\mathbf{1} = \mathbf{1}} \max_{e \in Q} \text{TV}(eT, F(e)) = \frac{1}{4}.$$

Proof. Write $u_i = T_{i1} \in [0, 1]$. At the four vertices $e^{ij} = (e_i + e_j)/2$, $i \in \{1, 2\}$, $j \in \{3, 4\}$, the desired first output probability is one for $i = 1$ and zero for $i = 2$. For fixed j , the realized probabilities differ by $(u_1 - u_2)/2 \leq 1/2$. Their two errors must therefore sum to at least $1/2$, and one is at least $1/4$. Taking $(u_1, u_2, u_3, u_4) = (1, 0, 1/2, 1/2)$ attains error $1/4$ at the vertices. Convexity bounds the error everywhere else by the vertex maximum. This also gives a finite strict witness for (9). $\square$

Definition 3.4 (Compatible stochastic lift). For the mean dynamics in (1), a lift of profile (k_t) consists of nonempty sections $Q_t = H_t \cap \Delta_{k_t}$, affine observable maps $\pi_t : Q_t \rightarrow D$, seed lifts $E_x \in Q_0$ with $\pi_0(E_x) = x/10$, and affine maps $F_{t,c} : Q_t \rightarrow Q_{t+1}$ with

$$\pi_{t+1} F_{t,c} = R_\alpha^c \pi_t.$$

Each $F_{t,c}$ must extend to an ambient row-stochastic matrix. For each final query j , the map $e \mapsto (1 + e_j^T \pi_N(e))/2$ must extend to a $[0, 1]$ -valued affine map on the whole simplex, equivalently to a binary stochastic decoder. The least possible peak coordinate count is denoted $\text{cw}_N(\alpha)$.

Theorem 3.5 (Operational meaning of a compatible lift). *The complete feasible coordinate profiles of compatible stochastic lifts are exactly the feasible hidden-state profiles of (1). In particular, $\text{cw}_N(\alpha) = W_N(\alpha)$. For supplied finite sections and supplied affine lift maps, all stochastic extension requirements, including the final decoder requirements, can be tested by Theorem 3.2.*

Proof. A machine supplies the sections and idle-suffix maps of Lemma 2.4. Restrict its actual transition matrices to these sections; their observable identity holds there. The final decoder already extends to the whole simplex. Conversely choose the stochastic extensions and start at the specified seed lift. Induction keeps the running distribution in Q_t and its observable image equals the desired rotated seed. The final extended decoder then gives exactly (1). The stochastic maps at different cuts are used together, not selected afresh after seeing a hidden parameter. Affinely spanning points suffice for each extension test; convexity ensures legality throughout a section. $\square$

The theorem defines the dynamic object and gives a local feasibility criterion when its geometry is fixed. It is not an efficient algorithm for searching over all lifts. Proposition 3.3 does not prove that every logarithmic extension of a regular polygon is incompatible with rotation; it proves that compatibility cannot be inferred merely from affine positivity on the section. In particular, we neither assume nor exclude a logarithmic hidden-state construction for (1). Static $O(\log m)$ extensions of regular m -gons [6, 12] remain relevant candidates, but need the inter-epoch and readout extension tests.

4. MEAN-SUPPORT GROWTH ALONG AN ORBIT

For a planar convex body P , let $h_P(u) = \max_{z \in P} u \cdot z$ and

$$m(P) = \frac{1}{2\pi} \int_0^{2\pi} h_P(\cos \theta, \sin \theta) d\theta.$$

For polygons this is perimeter divided by 2π , by Cauchy's perimeter formula. Define

$$\mathcal{G}_\alpha(P) = \text{conv}(P \cup R_\alpha P), \quad D_\alpha(P) = m(\mathcal{G}_\alpha(P)) - m(P).$$

The functional is monotone under inclusion and invariant under rotation. For a chain (7),

$$(11) \quad \sum_{t < N} D_\alpha(P_t) \leq m(P_N) - m(P_0) \leq \sqrt{2}.$$

The last bound uses $P_N \subset [-1, 1]^2 \subset \sqrt{2}\mathbb{B}^2$. It is deliberately larger than the exact endpoint allowance.

Lemma 4.1 (Accumulation without squaring). *For any integer $q \geq 2$ and P ,*

$$(12) \quad m\left(\text{conv} \bigcup_{k=0}^{q-1} R_\alpha^k P\right) - m(P) \leq (q-1)D_\alpha(P).$$

Proof. Write $h_k(\theta) = h_P(\theta - 2\pi k\alpha)$. Pointwise, $\max_{k < q} h_k - h_0 \leq \sum_{k=1}^{q-1} (h_k - h_{k-1})_+$. Each summand has the same integral as $(h_1 - h_0)_+$, whose normalized integral is $D_\alpha(P)$. Integrate the inequality. $\square$

Lemma 4.2 (Inscribed-polygon deficit). *Suppose $a\mathbb{B}^2 \subset P$, and P has at most $M \geq 3$ vertices. Put $r = \max_{z \in P} \|z\|_2$. Then*

$$(13) \quad m(P) \leq r \frac{M}{\pi} \sin \frac{\pi}{M}.$$

Proof. Push each vertex radially to the circle of radius r . The resulting convex hull still contains zero and hence contains the old vertices and P . If its successive angular gaps are $\theta_1, \dots, \theta_m$, $m \leq M$, its perimeter is $2r \sum_i \sin(\theta_i/2)$. Concavity of sine on $[0, \pi]$, $\sum_i \theta_i = 2\pi$, and monotonicity of $m \sin(\pi/m)$ give at most $2rM \sin(\pi/M)$. Monotonicity of mean support and Cauchy's formula prove the assertion. The radial hull contains zero because any positive affine dependence of the old vertices giving zero can be reweighted by their radii. $\square$

Lemma 4.3 (A denominator gives an orbit net). *If $(p, q) = 1$ and $|\alpha - p/q| < q^{-2}$, the first q points of the rotation orbit of any unit vector form an angular $3\pi/q$ -net.*

Proof. The multiples of p/q give the uniform grid of mesh $1/q$ turns. The corresponding first q multiples of α differ by less than $1/q$ turns. Every point is within $1/(2q)$ turns of the grid, proving the bound after multiplying by 2π . $\square$

Theorem 4.4 (One-step quantitative mean-support gain). *Suppose $a\mathbb{B}^2 \subset P$, $|\text{ext } P| \leq M$, $M \geq 3$, and a reduced fraction p/q satisfies $q \geq 8M$ and $|\alpha - p/q| < q^{-2}$. Then*

$$(14) \quad D_\alpha(P) \geq \frac{3a}{4M^2q}.$$

Proof. Take a vertex of maximum radius r . Its first q rotations and Lemma 4.3 imply

$$m\left(\text{conv} \bigcup_{k < q} R_\alpha^k P\right) \geq r \cos(3\pi/q).$$

Subtract (13). Since $M \geq 3$ and $q \geq 8M$, Taylor's inequalities for sine and cosine give

$$\begin{aligned} \cos(3\pi/q) - \frac{M}{\pi} \sin(\pi/M) &\geq \frac{\pi^2}{M^2} \left(\frac{1}{6} - \frac{\pi^2}{120M^2} - \frac{9M^2}{2q^2} \right) \\ &\geq \frac{9}{M^2} \left(\frac{1}{6} - \frac{10}{1080} - \frac{9}{128} \right) = \frac{301}{384M^2} > \frac{3}{4M^2}. \end{aligned}$$

The inequalities use only $9 < \pi^2 < 10$. Apply Lemma 4.1, use $r \geq a$ and $q - 1 < q$. The gain is linear in the integrated support deficit; an area cap is not introduced. $\square$

For an irrational $\alpha \in (0, 1)$, let p_k/q_k be its continued-fraction convergents, ignoring the initial repeated denominator one, and define

$$(15) \quad Q_\alpha(M) = \min\{q_k : q_k \geq 8M\}.$$

We use the elementary relations

$$(16) \quad (p_k, q_k) = 1, \quad |\alpha - p_k/q_k| < \frac{1}{q_k q_{k+1}} \leq \frac{1}{q_k^2}, \quad q_{k+1} = a_{k+1} q_k + q_{k-1}.$$

They follow by writing α with its remaining continued-fraction complete quotient and using $p_k q_{k-1} - p_{k-1} q_k = (-1)^{k-1}$; see also [9]. In particular, if $a_k \leq A$ for all $k \geq 1$, $Q_\alpha(M) < 8(A+1)M$.

Theorem 4.5 (Profile budget and bounded-type width). *Every exact hidden-state realization of (1) satisfies*

$$(17) \quad \sum_{t < N} \frac{1}{2^{2K_t} Q_\alpha(2^{K_t})} \leq \frac{40\sqrt{2}}{3}.$$

If $a_k \leq A$, then

$$(18) \quad \sum_{t < N} 2^{-3K_t} \leq 160(A + 1).$$

For vector-state realizations the same statements hold with 2^{K_t} replaced by K_t . Consequently $W_N \geq \frac{1}{3} \log_2(N/[160(A + 1)])$ and $V_N \geq (N/[160(A + 1)])^{1/3}$.

Proof. Use Lemma 2.4 and Theorem 4.4 with $a = 1/10$, $M = 2^{K_t}$, $q = Q_\alpha(M)$ in (11). This gives (17). Under the bounded-type assumption each gain is at least $3/[320(A + 1)M^3]$. The endpoint allowance gives

$$\sum_{t < N} M^{-3} \leq \frac{320(A + 1)\sqrt{2}}{3} < 160(A + 1).$$

In the vector-state class, use the other vertex bound in Lemma 2.4. A peak bound supplies the stated consequences. $\square$

For the golden angle all partial quotients equal one. The explicit hidden lower bound exceeds three when $N > 320 \cdot 2^9 = 163840$. The earlier area estimate exceeded three only after $24000000 \cdot 2^{18} = 6291456000000$ commands [16]. The new constant is still not a practically sharp finite-horizon classification. Its significance is the improved exponent in the profile budget and its matching vector-state construction below.

The same mean-support proof works in any dimension for a specified orthogonal command family once one supplies an orbit-covering estimate and an inscribed-polytope mean-support deficit. No higher-dimensional numerical exponent is asserted here. Unlike a volume argument, the potential is not tied to a determinant formula. A strictly contractive rotation instead has a finite invariant polygon; the divergent conclusion cannot simply be retained under contraction.

5. RESONANT POLYGONS AND EXACT SYNTHESIS

A regular polygon need not be enlarged by the worst possible angular misalignment. Choosing its order from a convergent changes the cost from quadratic to cubic in its number of vertices.

Lemma 5.1 (Resonant containment and stochastic rows). *Let $q \geq 5$, $(p, q) = 1$, and $|\alpha - p/q| < q^{-2}$. Put*

$$a_q = \pi/q, \quad \delta = 2\pi(\alpha - p/q), \quad \lambda = \frac{\cos(a_q - |\delta|)}{\cos a_q}, \quad w = \frac{\sin |\delta|}{\lambda \sin(2a_q)}.$$

Then $1 \leq \lambda$, $0 \leq w \leq 1$, and

$$(19) \quad \log \lambda \leq 40/q^3.$$

For $v_j = (\cos(2\pi j/q), \sin(2\pi j/q))$ and $P_t = r_t \text{conv}\{v_j : 0 \leq j < q\}$, $r_{t+1} = \lambda r_t$,

$$P_t \cup R_\alpha P_t \subset P_{t+1}.$$

The two stochastic rows realizing the inclusions are

$$(20) \quad T_0(j, \cdot) = \lambda^{-1} \delta_j + (1 - \lambda^{-1}) \text{Unif}[q],$$

$$(21) \quad T_1(j, \cdot) = (1 - w) \delta_{j+p} + w \delta_{j+p+\text{sgn } \delta},$$

with indices modulo q ; when $\delta = 0$, take the pure shift.

Proof. Here $|\delta| < 2\pi/q^2 \leq a_q$. The point $r_t v_{j+p}$ rotated by the residual angle δ lies on the edge of P_{t+1} joining the two displayed vertices. The imaginary coordinate gives the coefficient w ; the real coordinate identity is

$$\lambda = \cos |\delta| + \tan(a_q) \sin |\delta|.$$

This proves both barycentric equality and $0 \leq w \leq 1$ by edge containment. The idle row uses the zero barycenter of a regular polygon. Finally,

$$\log \lambda = \int_{a_q - |\delta|}^{a_q} \tan u \, du \leq |\delta| \tan a_q \leq (2\pi/q^2)(2\pi/q) < 40/q^3.$$

We used $a_q \leq \pi/5 < \pi/4$, where $\tan u \leq 2u$. $\square$

Theorem 5.2 (Cube-root synthesis). *Let q_N be the first convergent denominator of α at least $\max\{5, (40N)^{1/3}\}$. There is an exact vector-state realization of (1) with q_N states at every command cut. When the partial quotients are bounded by A ,*

$$q_N < (A+1) \max\{5, (40N)^{1/3}\}, \quad V_N(\alpha) = \Theta_\alpha(N^{1/3}).$$

The command matrices can be the same at all epochs. The final decoder and the chosen polygon order are horizon-specific.

Proof. Take $q = q_N$, $r_0 = 1/4$ and $r_t = r_0 \lambda^t$. The inradius of P_0 is at least $\cos(\pi/5)/4 > \sqrt{2}/10$, so all seed vectors lie in P_0 . By (19), $r_N \leq e/4 < 3/4$, hence all polygons lie inside D . Initialize by any convex representation of $x/10$ in P_0 , and use (20)–(21). Given final label s and query j , use binary mean $r_N e_j^T v_s$. The row identities imply by induction that the distribution's barycenter in P_t is exactly the rotated seed. This proves every conditional row of (1), not an approximation to the angle. For an explicit initialization, the ray through x meets one edge of P_0 ; represent that boundary point by its two endpoints and mix with the uniform zero barycenter to obtain $x/10$. The continued-fraction recurrence bounds the first crossing denominator by $(A+1)$ times the threshold. Theorem 4.5 supplies the matching vector-state lower order. $\square$

This proves (2)–(3). In the normalized label coordinates, both command matrices contract the observable barycenter by λ^{-1} at each command. Multiplying the final decoder by r_N compensates for exactly N contractions. Querying that same decoder earlier generally gives the wrong amplitude. Thus the stationary command rows in this construction are not a clock-free anytime realization, and do not contradict Section 7.

The construction uses direct vertex states; it does not exploit the logarithmic extensions of regular polygons. The matched order for V_N is a theorem about all vector-state chains, not evidence that $W_N = V_N$. The gap $\Omega_\alpha(\log N) \leq W_N \leq O_\alpha(N^{1/3})$ remains the quantitative hidden-lift question. The new upper bound improves the previous $O(N^{1/2})$ construction without asserting its own optimality in the unrestricted class.

Corollary 5.3 (A contractive comparison). *Replace the active mean map by ρR_α , $0 < \rho < 1$, and keep the idle map. A single horizon-independent vector-state device realizes all command lengths with any convergent order $q \geq \max\{5, (40/(-\log \rho))^{1/3}\}$. For bounded-type angles this uses $O_\alpha(1 + (-\log \rho)^{-1/3})$ processing labels.*

Proof. Use one fixed polygon of radius $1/4$. The resonant estimate implies $\rho \lambda \leq 1$, so $\rho R_\alpha P \subset P$ and the idle map preserves P . The active row is the boundary row of (21) mixed with uniform mass, with weight $\rho \lambda$ on that row. A fixed decoder reads the same polygon at all times. This is a quantitative instance of the classical invariant polytope construction [3], not a new existence principle. $\square$

6. AN EXPLICIT RATE FOR A RATIONAL ROTATION

Let $\zeta = (3 + 4i)/5 = e^{2\pi i\alpha}$, $0 < \alpha < 1/4$. This is not a root of unity: otherwise $\zeta + \zeta^{-1} = 6/5$ would be a rational algebraic integer and hence an integer. Thus α is irrational. No bounded-type assertion about this angle is needed.

Lemma 6.1 (Elementary denominator control). *For every integer $r \geq 1$,*

$$\|r\alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{5^{-r}}{2\pi}.$$

Consequently the denominator function of (15) satisfies $Q_\alpha(M) < 7 \cdot 5^{8M}$ for every integer $M \geq 3$.

Proof. The Gaussian integer $(3 + 4i)^r - 5^r$ is nonzero, so its modulus is at least one. Therefore

$$5^{-r} \leq |\zeta^r - 1| \leq 2\pi \|r\alpha\|_{\mathbb{R}/\mathbb{Z}}.$$

If $q = Q_\alpha(M)$ and r is its preceding convergent denominator, then $r < 8M$ and (16) gives $\|r\alpha\| < 1/q$. Combine the two estimates and use $2\pi < 7$. The elementary bound is exponentially weak in M , but entirely explicit. $\square$

Theorem 6.2 (Rational-angle profile bounds). *For the rotation $R = (1/5) \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}$, every exact hidden-state profile obeys*

$$(22) \quad \sum_{t < N} 2^{-2K_t} 5^{-8 \cdot 2^{K_t}} \leq 140.$$

In particular,

$$(23) \quad N \leq 140 \cdot 2^{2W_N} 5^{8 \cdot 2^{W_N}}.$$

For $N > 140$ this implies

$$W_N \geq \max \left\{ 3, \log_2 \left(\frac{1}{9} \log_5 \frac{N}{140} \right) \right\}.$$

In the vector-state class, $\sum_{t < N} K_t^{-2} 5^{-8K_t} \leq 140$ and, for $N > 140$, $V_N \geq \max\{3, \frac{1}{9} \log_5(N/140)\}$. Both classes have an exact rational realization with peak $3(N+1)$.

Proof. Insert Lemma 6.1 into (17). It gives

$$\sum_{t < N} M_t^{-2} 5^{-8M_t} \leq \frac{280\sqrt{2}}{3} < 140, \quad M_t = 2^{K_t}.$$

Each summand is decreasing in M_t , proving (23). For an integer $W \geq 1$, $2W \log_5 2 \leq 2^W$; hence $2 \log_5(2^W) + 8 \cdot 2^W \leq 9 \cdot 2^W$. Taking logarithms gives the stated hidden lower bound. For vector states use $M_t = K_t$ and $2 \log_5 V \leq V$ for integers $V \geq 1$.

For completeness, an exact rational upper machine stores (s, k) , where s is one of the three triangle vertices in Proposition 2.3 and k is the number of active commands. Initialize s by the rational barycentric coordinates of $x/10$. An idle command leaves (s, k) unchanged, and an active command increments k . At epoch t only $0 \leq k \leq t$ is needed. Assign predictive vector $R^k v_s$ to (s, k) and use its coordinate as the final binary mean. All assigned vectors have norm less than one, all transitions are rational, and the row identities hold on every state. The largest processing alphabet is $3(N+1)$; a held binary answer needs only two labels in its separate terminal cut. $\square$

The explicit inequalities replace the former quantifier-elimination-only exclusion for this angle. They do not assert its sharp asymptotic order. Sharper irrationality estimates for α would improve the denominator function in the same profile theorem, but none is silently assumed here. The rational upper machine admits the separately priced fair-bit implementation of [15]; its atomic label optimum and its sampler workspace are different quantities.

7. REMOVING THE CLOCK: ONE DECODER AT EVERY STOPPING LENGTH

We now change the query-time specification explicitly. The commands still come from the same external alphabet, and there is still only one terminal query. It may, however, arrive after any number $0 \leq n \leq N$ of commands, and the answer must have mean $e_j^T R_\alpha^{\sum c_t} x/10$ at that length. The machine does not receive n as a separate input.

Definition 7.1 (Anytime autonomous realization). An anytime autonomous machine has one finite set of available states, fixed command matrices T_0, T_1 , fixed initialization rows, and one fixed decoder for each query. The same rows work for every command length from zero through N . Let $A_N(\alpha)$ be the least total number of states, counting any states used to retain the emitted answer. The device may depend on the prescribed maximum N ; its rows may not depend on the actual stopping length. Tables and exact atomic sampling remain free, so this is not a succinct bit-space model.

Lemma 7.2 (Finite positive peripheral spectrum). *Every eigenvalue of modulus one of a finite row-stochastic matrix is a root of unity.*

Proof. Let $Tv = \lambda v$ and choose i with $|v_i|$ maximal and positive. Equality in the triangle inequality for $\lambda v_i = \sum_j T_{ij} v_j$ forces $v_j = \lambda v_i$ for every positive successor of i . All these successors have the same maximal modulus. Iteration follows a positive directed path, which eventually contains a cycle of length $l \geq 1$. Along that cycle, $v_j = \lambda^l v_j$ with $v_j \neq 0$. Thus $\lambda^l = 1$. This is the familiar peripheral-spectrum constraint for positive realizations; see [2, 3]. $\square$

Theorem 7.3 (Linear autonomous state complexity). *For any infinite-order planar rotation and every $N \geq 1$,*

$$(24) \quad N + 1 \leq A_N(\alpha) \leq 3(N + 1) + 2.$$

The upper machine has rational rows when R_α is rational. Thus $A_N(\alpha) = \Theta(N)$, whereas the fixed-length vector-state width is $\Theta_\alpha(N^{1/3})$ for bounded-type angles.

Proof. Suppose a machine has $K \leq N$ total states. Fix one nonzero seed and use only active commands. Let $d = d_1 + id_2$ be the complex column formed from its two fixed terminal means and let $z \neq 0$ be the initial complex predictive vector. With $T = T_1$ and $\zeta = e^{2\pi i \alpha}$, correctness says

$$ET^j d = z\zeta^j \quad (0 \leq j \leq N).$$

Write the characteristic polynomial of T as $\chi(X) = \sum_{j=0}^K a_j X^j$. Cayley–Hamilton and the first $K + 1$ required lengths give

$$0 = E\chi(T)d = z \sum_{j=0}^K a_j \zeta^j = z\chi(\zeta).$$

Thus ζ is an eigenvalue of T , contradicting Lemma 7.2. Hence $K \geq N + 1$. The argument needs only the finite length interval, not an infinite stationary specification or a minimality assumption.

For the upper bound use the labels (s, k) , $1 \leq s \leq 3$, $0 \leq k \leq N$, from the triangle-counter construction in Theorem 6.2, now all available in one state set. The idle matrix is the identity; the active matrix increments k , with arbitrary normalized rows after $k = N$. The decoder at (s, k) reads $R_\alpha^k v_s$ and has no epoch argument. Its norm is less than one. Initialization is the seed's triangle decomposition. The same matrices and decoder therefore answer at every permitted length. Add two absorbing answer states if the output must persist after its emission. This proves the count and rationality. $\square$

Equation (24) completes the proof of Theorem 1.1. It does not lower-bound an autonomous device required to answer only at one fixed length and allowed to be incorrect at all earlier lengths. That is another problem. In particular, the resonant construction in Section 5 already has stationary command rows, but its decoder compensates for exactly N normalizing contractions. The strengthened anytime requirement is what prevents that compensation from concealing the elapsed length. The finite-horizon comparison is a concrete use of classical spectral constraints, not a claim to discover the stationary rotation obstruction.

8. CLASSICAL COMPARISONS AND THE REMAINING LIFT PROBLEM

8.1. Fixed degree and positive order. The direct precedent for fixed degree three with unbounded positive order is Benvenuti–Farina [1], not merely the familiar stationary irrational-rotation obstruction. In the normalization reproduced by Czaja–Jaming–Matolcsi [5, Example 2, equation (3.1)], their family is

$$H_m(z) = \frac{1}{z-1} - \frac{4(5/2)^{m-2}}{z-2/5} + \frac{3 \cdot 5^{m-2}}{z-1/5}, \quad m \geq 4.$$

Its McMillan degree is three. The 1999 lower bound on positive order is m , and the later construction attains m [1, 5]. Those statements concern one repeated nonnegative matrix and an entire impulse response. They are not lower bounds against independently chosen stochastic matrices at every cut of a finite word experiment.

There is an equally direct precedent for the inadequacy of positive factorization without dynamics. Benvenuti–Farina [2, Section 3, Example 4 and Theorem 2] give a periodic Hankel matrix with a nonnegative factorization of inner size three but no three-dimensional positive realization; their minimality theorem includes a positive intertwining matrix. Thus neither the need for compatibility nor its failure in positive factorization is claimed as new here. For a finite prefix–suffix array, conditioning supplies a positive factorization at a cut, just as for a finite Hankel truncation. The issue in this article is one sequence of such factorizations under every command, allowing unrelated rows and a horizon-dependent machine. The profile budget proves a necessary inequality even in that enlarged class. It is not claimed to sharpen the known stationary lower bound for the family H_m .

Accessible spaces, observable quotients and invariant cones are classical [8, 3, 13, 10]. Lemma 2.4 is the finite-word version needed to avoid asserting an intertwining identity on hidden directions that the observations do not determine. The proof is included to expose that quantifier, not to claim invention of an accessible-space construction.

8.2. Static and dynamic extensions. Yannakakis’ factorization theorem relates extension size to nonnegative rank of a slack matrix [14]; see the explicit formulation in [6, Theorem 1]. The number of faces of an extension supplies the standard logarithmic lower estimate from projected vertices. Fiorini–Rothvoß–Tiwary [6, Theorem 2], building on Ben-Tal–Nemirovski [4], show that a regular m -gon has an $O(\log m)$ extension. Vandaele–Gillis–Glineur [12] refine regular-polygon bounds using slack factorizations.

These theorems minimize a static inequality count. They do not require the lifted command maps or the observable decoder to extend to stochastic maps on the whole ambient simplex. Proposition 3.1 shows that bounded polytope extensions can be normalized without increasing their facet count; Proposition 3.3 shows why this normalization alone does not settle dynamical implementation. The extension test (9) is elementary convex separation. Its role is to make the missing condition exact and checkable for fixed lift data, not to claim a new general linear-programming theorem.

The substantive quantitative conclusion is therefore divided in two. For arbitrary hidden realizations, the new potential yields a logarithmic lower bound, while the resonant construction yields a cube-root upper bound. For vector-state realizations, the same potential has no exponential facet-to-vertex conversion, and the order is exactly $N^{1/3}$. An independent small static lift does not bridge this gap. Conversely, the map-defect example does not rule out small compatible lifts that have a different geometry.

8.3. What is retained, and what is not inferred. The older volume obstruction, rank-three two-cut example, exact tagged profile identity, online query constructions and physical saddle results remain in their original sources and in the supporting archives [15, 16]. In particular the tagged identity still requires exact final tag recovery and a common suffix interface. Nothing in the present paper extends it to noisy tags or overlapping supports. The earlier additive replacement update is a weighted reservoir construction; its sampling mechanism is credited to that literature [11, Appendix A.2], not used as a new claim here.

The finite proofs above are self-contained apart from the named classical background identities whose short arguments are also given. The source record distinguishes the original 1999 reference from the full author survey and later matching-order proof used to inspect its family. The original 1999 article and the original Yannakakis article have not been claimed to receive an exhaustive proof-by-proof priority audit. Their results are expressly credited; the full accessible comparisons [2, 5, 6, 12] supply the concrete theorem statements used here. No exhaustive originality clearance is inferred from this bibliography.

The main unresolved issue is the optimal hidden width between logarithmic and cube-root growth. The vector-state result is not substituted for that question. The rational-angle bound is explicit but much weaker; its sharp denominator-controlled rate is likewise not identified.

The autonomous theorem concerns one fixed decoder at all stopping lengths, not just removal of a clock from the original single-length specification. All counts continue to price persistent labels rather than read-only programs, arithmetic, or physical realization of irrational coins. These are different optimization problems, and the comparisons depend on preserving their definitions.

The program title identifies this manuscript within the preserved research series. It does not assert that finite positive-realization arguments replace independent Fourier, large-deviation, semigroup, or operator-domain proof obligations elsewhere in that series, nor does it assert a journal decision. The theorem chain established here is the one displayed in Theorem 1.1 and its supporting lift and profile statements.

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